HARNESSING THE REPLICATION EUCATASTROPHE: BLOCKCHAIN AND CRYPTOCURRENCIES AS LEVERS FOR OPEN SCIENCE

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ABSTRACT

The **replication crisis** may be reframed as a ***Replication Eucatastrophe***: a disruptive yet ultimately **constructive turning point** that catalysed **reform** toward greater scientific credibility, transparency, and reproducibility. This opportunity has been taken up by the **Open Science movement**, whose trajectory is articulated in **Nosek's (2019)** ***Strategy for Culture Change***: openness must first become possible, then easy, normative, rewarding, and ultimately required. Despite substantial progress, however, many **Open Science (OS) practices** remain **unevenly adopted, weakly enforced**, and constrained by **distorted incentive** structures. Against this backdrop, I propose a **blockchain-based framework** designed to **operationalise and strengthen OS** by embedding its principles directly into the research lifecycle. Integrated with existing solutions such as the **Open Science Framework** and **Open Badges**, blockchain technologies and cryptocurrencies can function as **levers for OS** on multiple levels by enabling immutable and verifiable records, certified credentials, incentive mechanisms and decentralised governance.

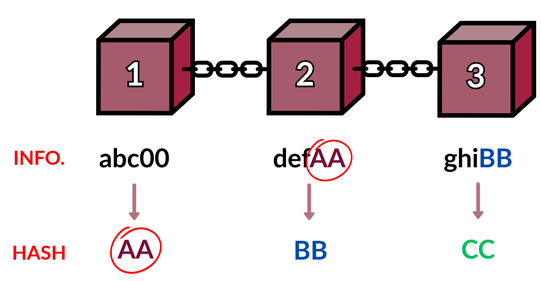
HOW TO OPEN SCIENCE: A STRATEGY FOR CULTURE CHANGE

To challenge the status quo and guide academia toward an **Open Science paradigm** [1], Nosek's ***Strategy for Culture Change*** [2] aims to **engage all scientific stakeholders** through five sequential steps:

1. **MAKE IT POSSIBLE** – Highly motivated innovators need only **adequate infrastructure** and **tools** to implement OS practices;
2. **MAKE IT EASY** – Early adopters demand **user-friendly solutions**;
3. **MAKE IT NORMATIVE** – The **visibility** of early adopters' behaviour is crucial for the **normative influence** they exert on peers;
4. **MAKE IT REWARDING** – Real behavioral change necessitates **reforms** in the academic **incentive** system that make it **desirable**.
5. **MAKE IT REQUIRED** – Structural change also requires **new policies mandating open practices.**

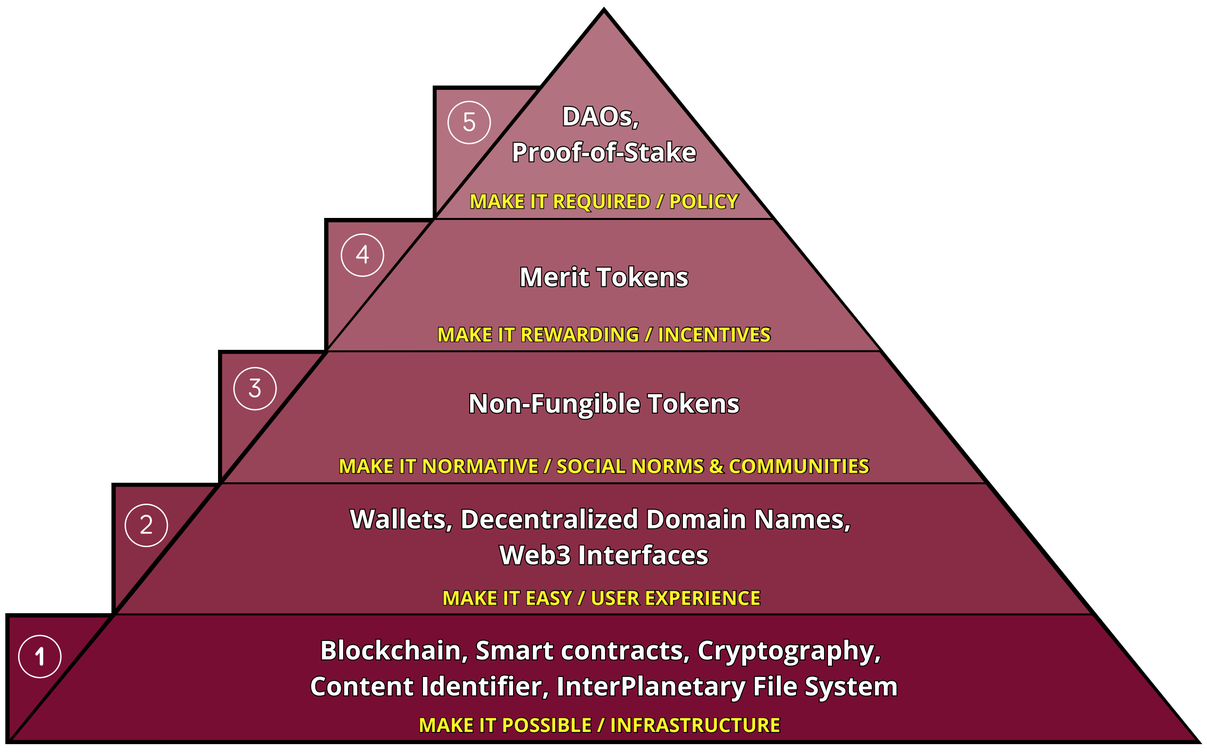
Despite advances in OS initiatives, a **substantial adoption gap** persists [3;4;5], which technology may help address [6]. I therefore propose a **blockchain-based framework** to **complement existing OS strategies**, drive cultural change **across all levels of Nosek's pyramid**, and support the entire **research lifecycle**, while remaining adaptable to other disciplines.

ESSENTIALS OF BLOCKCHAIN TECHNOLOGY



Blockchain is a **Distributed Ledger Technology** [7;16] that enables **secure, transparent, and immutable** recording of information in a "digital registry" distributed across a **peer-to-peer network** of computers (nodes), each maintaining a copy of the entire blockchain. **Data is stored in blocks** that are **cryptographically linked**, forming a chain, while nodes participate in **validating** these blocks. Blockchain underpins the functioning of **cryptocurrencies** such as Bitcoin.

BLOCKCHAIN CONTRIBUTIONS TO THE *CULTURE CHANGE* PYRAMID



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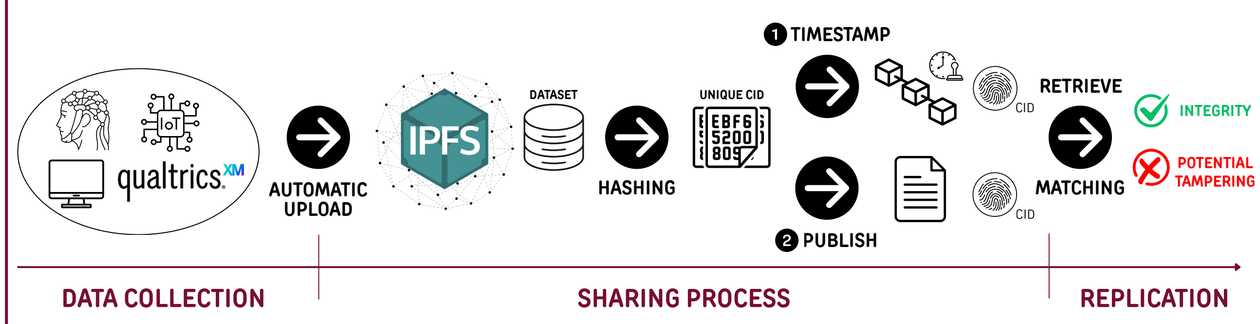
BLOCKCHAIN FOR OPEN SCIENCE: STEP BY STEP

I will illustrate how the proposed framework can **advance Open Science** by presenting **blockchain-based solution** for the 1st, 3rd and 4th stages of **Nosek's pyramid**, while recognizing that its applicability in academia extends beyond the examples provided [17].

OPENING DATA WITH BLOCKCHAIN & IPFS

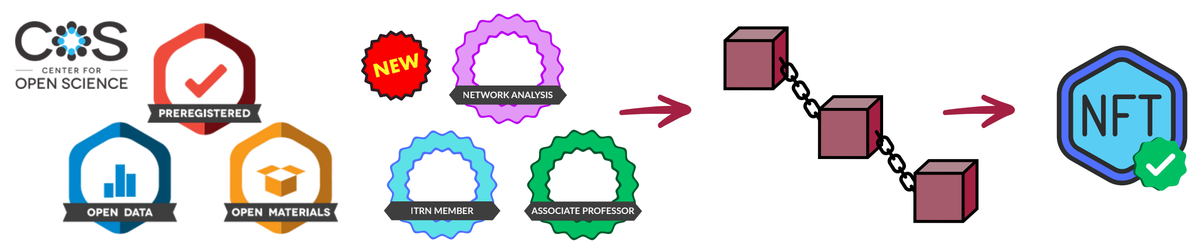
To make OS possible, the **foundational infrastructure** could be reinforced with blockchain features. In particular, the open sharing of data, materials, and methods could be enhanced through blockchain-based **time-stamping** [8] and the **decentralized storage** provided by the **InterPlanetary File System (IPFS)** [9]. To illustrate this process, I will outline how IPFS and blockchain could **facilitate data sharing:**

1. A researcher **collects data** through laboratory instruments, online survey platforms, or Internet of Things devices;
2. The raw data is **automatically uploaded** to **IPFS** for decentralized storage, thus **preventing** potential **manipulation** by the researcher;
3. IPFS assigns each dataset a unique **Content Identifier (CID)**, a hexadecimal string generated via cryptographic hashing [10]. The CID is intrinsically **tied to the file's content**, so that even minor modifications produce a different identifier;
4. The CID is **recorded** in a specific **blockchain block**, thereby creating a **tamper-proof timestamp** that certifies the dataset's existence at a given moment with that exact content;
5. The researcher publishes the results, including the **block identifier** and **CID** to enable **retrieval** of the raw dataset from IPFS and the **verification of its integrity**;
6. To verify that the open raw dataset used in a study **corresponds** to the one **timestamped** on the blockchain, the associated **CIDs** are **compared**: a match indicates that the underlying data are **consistent** and untampered, whereas a mismatch signals potential **manipulation**;
7. Anyone can **retrieve** the raw data from IPFS using the CID and attempt to **reproduce** the results. However, **encrypting** the CID prior to its blockchain time-stamping still allows verification of the information's integrity **without immediate disclosure** to the scientific community.



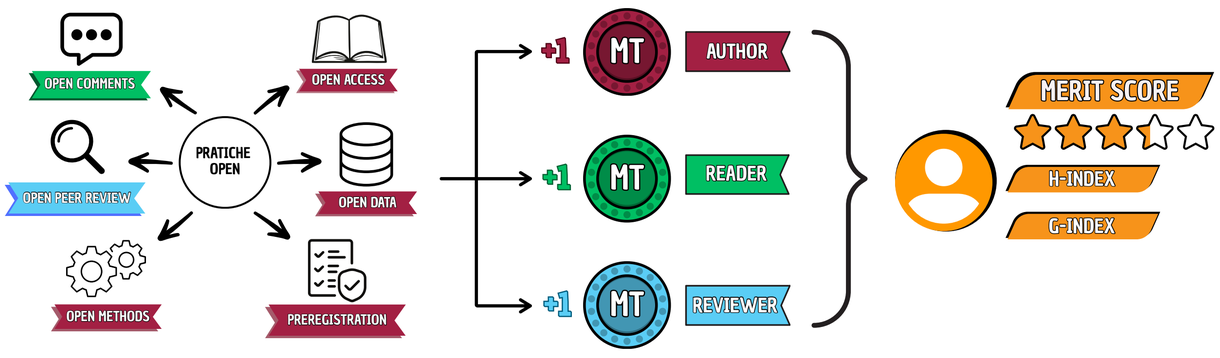
NFTs: CERTIFIED CREDENTIALS

To make OS normative, **Open Badges** [11] can be issued as **Non-Fungible Tokens** (NFTs) [16]: digital credentials anchored on the blockchain that can denote **qualifications**, academic **achievements**, **intellectual property**, skills, areas of **expertise**, and **adherence to OS practices**. NFTs are **resistant to forgery**: validated and immutably recorded by blockchain consensus, their **authenticity is verifiable** without involving third parties.



MERIT TOKENS: INCENTIVES FOR OPEN SCIENCE

To make OS rewarding, blockchain and cryptocurrencies can also establish a novel **incentive system** for scientific research. **Merit Tokens (MTs)** could be **awarded** to stakeholders who perform virtuous actions aligned with **OS principles**. [12]. MTs can be earned in various roles (such as Author, Reviewer, or Reader) and an equation could yield a **Merit Score (MS)**, reflecting an individual's impact on science. MTs could incentivize honest, high-quality **peer reviews** [13]. Additionally, MS could serve as a qualitative, decentralized **measure of academic reputation** grounded in OS principles, complementing traditional quantitative scientometrics. The **selection of optimal reviewers** for the **OPR** could rely on their **MS**, complemented by certified skills and areas of expertise validated through **NFTs** [14].



CONCLUSIONS

This blockchain-based framework aims to enhance **transparency, fairness**, and **incentive alignment**, fostering a more robust and reproducible scientific ecosystem. Its **feasibility, benefits, and effectiveness** require further study, alongside **critical technical aspects** such as blockchain architecture, consensus mechanism, and tokenomics design.